S&P GLOBAL × CRISIL CAMPUS HACKATHON 2026 · AI/NLP RISK ENGINE CASE STUDY

Sentinel.

A keyless, multilingual **AI/NLP Risk Engine** that turns news and social text into **sentiment, event and impact signals**, then drives **both** downstream modules: a sentiment-tilted index (A) and a bank-book stress test (B).

9
keyless sources: news, filings, central banks, social. **0 API keys**

0.85
sentiment macro-F1 on an uncontaminated test set (lexicon baseline: 0.45)

23
stress tests fired automatically on the replayed 24 Feb 2022 news flow

0.21 s
10,000-path Monte Carlo on a $10 bn, 300-position bank book

Yugh Juneja · VIT

github.com/yugh88/VIT_YughJuneja_Hackathon · runs offline with `docker compose up`

PROBLEM & APPROACH

Market-moving risk arrives as text, in many languages, faster than people can read it.

The business problem

A risk desk needs to answer four questions, minutes after a story breaks:

1. **Who** is affected? Which company or country
2. **How** does the news read? Sentiment −1 … +1
3. **What** kind of event is it? Credit, geopolitical, M&A …
4. **How big** could it be, and **what does it do to our book?**

Lexicons and keyword alerts miss context, ignore non-English sources and cannot size the hit. A single headline should never move capital on its own.

1 **Ingest everything, never go dark**
9 keyless sources (GDELT in 65 languages, RSS, SEC 8-K, Fed/ECB/RBI, Mastodon, Reddit, HN, tweets). Circuit breakers fail over to recorded replays.

2 **Read like an analyst**
Fine-tuned FinBERT + XLM-R score sentiment *per entity*; 12 event classes; 284 entities with 9.7k aliases in 20+ languages.

3 **Learn impact from the market, not from rules**
The Impact Score (1–10) is trained on 74,964 past news items labelled with the abnormal stock or index move that followed.

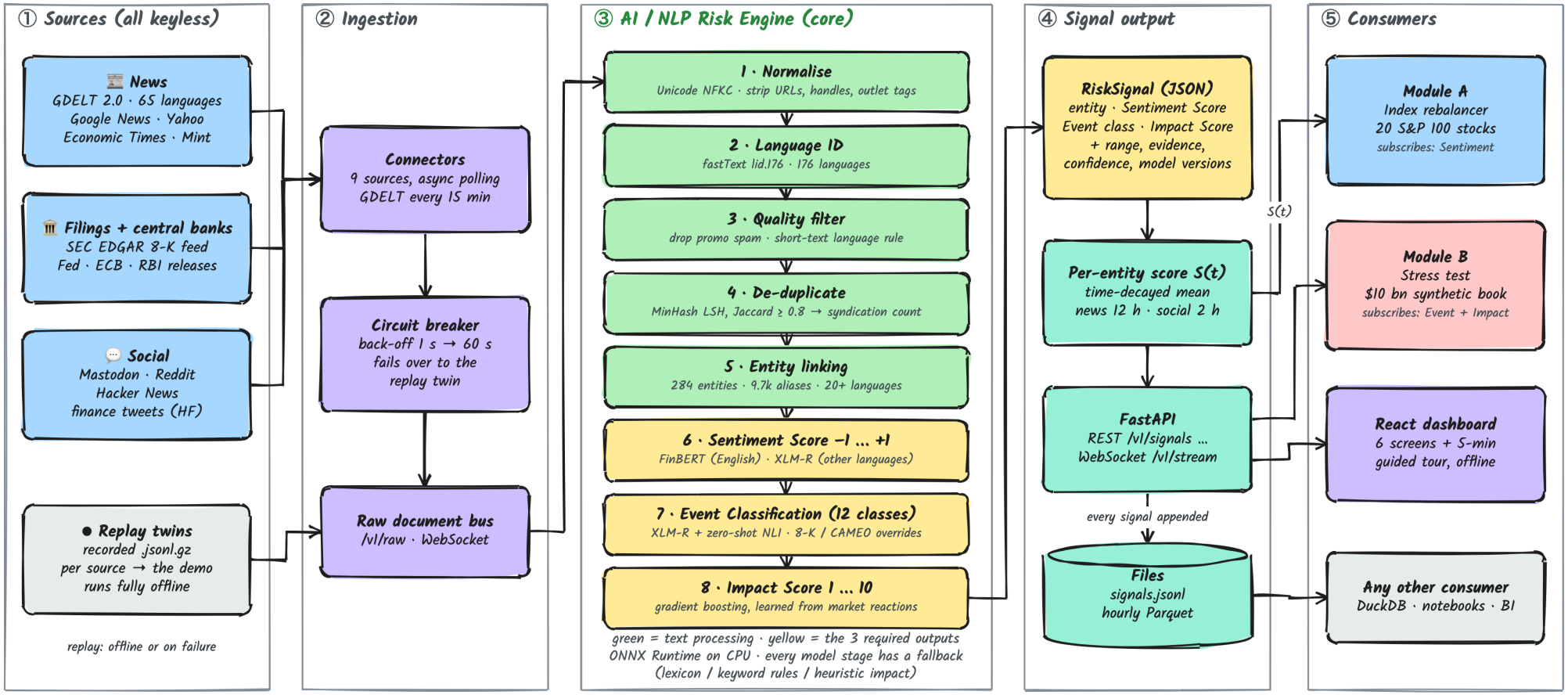
4 **Act on it, and test honestly**
Both modules built: **A** index rebalancer and **B** stress test. Time-split tests, frozen sets, placebos, and every miss published.

THE BRIEF → WHAT WE BUILT

✓ ≥ 2 sources	✓ Sentiment −1…1	✓ Event class	✓ Impact 1–10	✓ API or file	✓ ≥ 1 module
9 sources, news + social	per entity, calibrated	12 classes, 5 required ones in	learned, with range + drivers	REST, WebSocket, JSONL, Parquet	both A and B, with dashboards

SYSTEM DESIGN

Five stages, one structured signal, three consumers



Output via REST · WebSocket · JSONL · Parquet. More diagrams in docs/diagrams/: data flow, Module A, Module B, impact training, wireframe

IMPLEMENTATION HIGHLIGHTS

Each design choice is backed by a measured result

CHOICE	WHY	EVIDENCE (MEASURED)
Fine-tuned FinBERT (EN) + XLM-R (multilingual), entity-aware input	Lexicons miss finance context and which company a sentence is about	**Macro-F1 0.85 vs 0.45 for LM + VADER (+0.40)**
Impact = gradient boosting on realised abnormal returns	Hand-written severity rules are not grounded in market outcomes	**Spearman ρ 0.115 vs 0.043 heuristic (2.7×), 2024+ out of sample**
Cheap filters before any model: entity gate, MinHash dedupe, spam rules	Most documents mention no tracked name or are wire copies	**≥ 64% of docs skip the models; 13% collapsed as duplicates**
ONNX Runtime FP32 with fp16 storage, not INT8	Full INT8 cost 2–3 accuracy points on our data	**Model files halved; prediction agreement 1.00**
Corroborated trigger: > 7 impact, ≥ 3 outlets incl. tier-1, 6 h clusters	One headline should not move capital	**24 Feb 2022: 3,107 signals → 23 tests (135× fewer)**
Shocks calibrated on real episodes; Basel IRB + Monte Carlo	Defensible numbers instead of an arbitrary "–10%"	**Ukraine 2022 · 2022 hikes · SVB 2023; 10k draws in 0.21 s**
Circuit breakers + replay twins for every source	Free sources rate-limit and fail	**Demo and tests run with zero network access**

2 / 2
downstream modules built (brief asks for ≥ 1)

12.4k
lines of Python: engine, modules, training

6
dashboard screens + guided tour (React, TypeScript)

45
automated tests · lint · key scan in CI

83
logged design decisions with evidence

Python 3.11 · FastAPI · ONNX Runtime · Hugging Face · scikit-learn · DuckDB/Parquet · React + TypeScript + ECharts · Docker · GitHub Actions

KEY RESULTS · HELD-OUT TEST DATA ONLY

Reads finance text well; impact scores track real moves

0.851

Sentiment macro-F1

Twitter fin-news test, n = 2,388 · ECE 0.030 · lexicon 0.452

✓ target ≥ 0.85

0.899

Entity-level sentiment

SEntFiN held-out, n = 784 · lexicon 0.512

✓ +0.39 over baseline

0.818

Event macro-F1

9 classes, n = 4,117 · train/test near-duplicates removed

✓ target ≥ 0.80

100%

Credit-event recall

150 hard 2024+ headlines · F1 0.80 · precision 0.67

✓ F1 target ≥ 0.65

1.61×

Lift at Impact ≥ 8

share of top-decile real moves vs chance (16.8% vs 10.4%)

30,153 out-of-sample items

≥ 64%

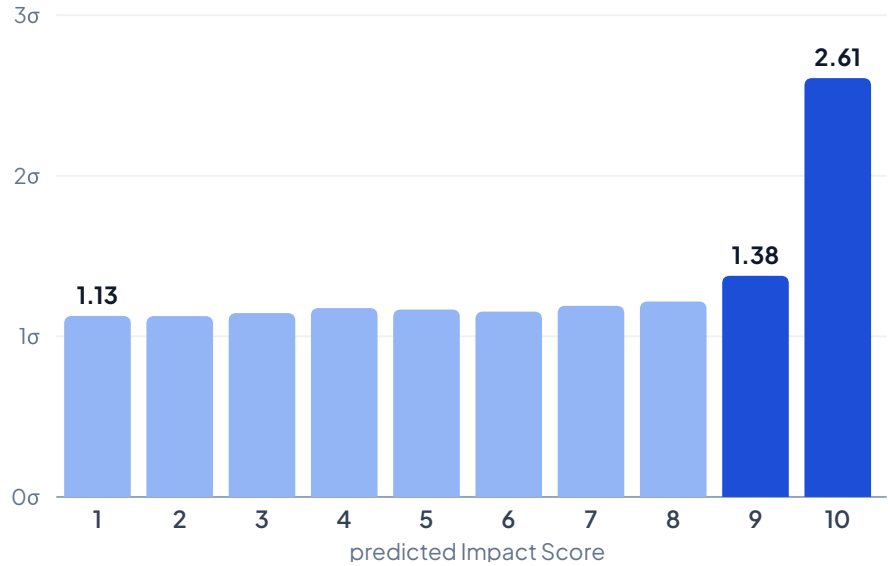
Docs skip the models

no tracked entity or near-duplicate → exit before inference

efficiency vs. naive "score all"

Higher Impact Score → bigger real moves

Mean realised move (in σ of the stock's own residual volatility) by predicted score · 2024+ test

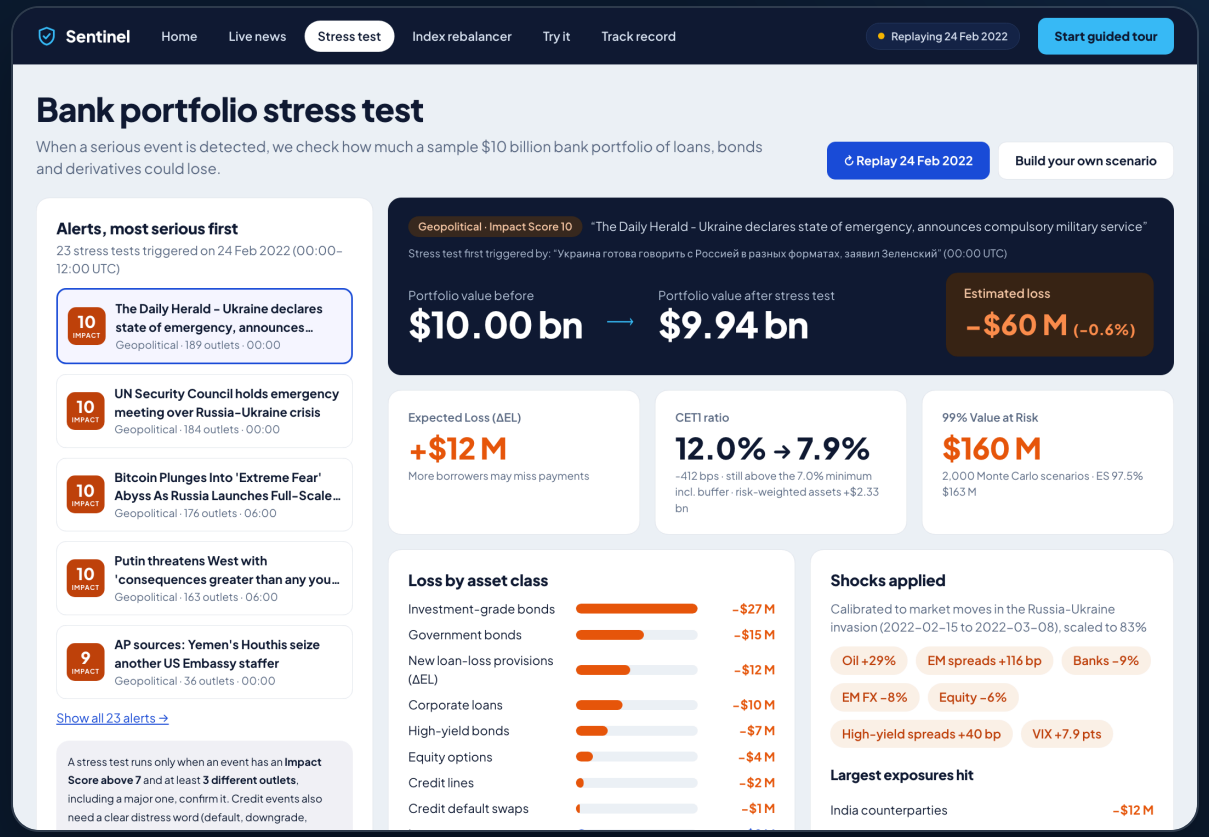


Score 10 moves 2.3× more than score 1. Scores 1–8 are nearly flat (an honest limitation).

Every number is regenerated by **make eval / make bench / make backtest** → docs/model_card.md

DOMAIN IMPACT · REPLAY OF 24 FEB 2022 (REAL RECORDED NEWS, REAL MODELS)

From raw headlines to a capital number for the risk committee, automatically



11,520 articles · 00:00–12:00 UTC

3,107 risk signals linked to a company or country

23 tests corroborated, merged per 6 h cluster

8 / 9 top-ranked alerts are war clusters

- **Early, multilingual:** the first tests fired at 00:00 UTC from **Russian-language** headlines.
- **Sized, not just flagged:** each alert reports loss, ΔEL, ΔRWA, ΔCET1, 99% VaR and 97.5% ES. Example: Russia geopolitical, –$72 m and –437 bp CET1.
- **Explainable:** every score carries its evidence sentence, drivers and a likely range, which suits model-risk review.
- **Cheap to run:** keyless, CPU only, works offline. Module A shows the same signal steering an index within hard limits.

Module B also verified on scenarios: Macro (US, impact 8) –$321 m · $10.0 bn → $9.68 bn · ΔCET1 –743 bp

LIMITATIONS & NEXT STEPS · WHAT DID NOT WORK, STATED PLAINLY

We met the core accuracy targets and published every miss

TARGET	GOAL	MEASURED	
English sentiment macro-F1	≥ 0.85	0.851	✓ met
Event macro-F1	≥ 0.80	0.818	✓ met
Credit-event F1 / precision	0.65 / 0.80	0.80 / 0.67	½ partly
Native multilingual sentiment F1	≥ 0.75	0.53	✕ missed
Impact Spearman ρ	≥ 0.25	0.115	✕ missed
Latency p95 / throughput (CPU)	150 ms / 150 docs/s	254 ms / 10–13	✕ missed
Module A beats 95% of placebos	net IR > 0	IR −1.43 · 33%	✕ no edge

Assumptions

Synthetic seeded bank book (sector mix from the public transactions dataset). Evaluation sets annotated blind and in two passes, with a human spot-check pending. Recorded tweets stand in for X, which has no keyless API.

Next steps

1. Human-labelled native-language data to close the 0.53 → 0.75 gap (the translated proxy scored 0.83 and overstated it)
2. Static INT8 or GPU workers behind an accuracy gate (≈ 3.5×); 10 M docs/day needs about 90–116 CPU workers today
3. Intraday impact labels; more hard negatives for credit precision
4. Plug real exposure feeds in place of the synthetic book

🔍 **Why publish the misses?** A machine-translated test said our multilingual model scored 0.83; real native headlines said 0.53. We report the real number. Every metric comes from a time-split or frozen test set, touched once, and is regenerated by a script.

Details: docs/model_card.md · decision log docs/DECISIONS.md